In [34]:
```
!pip install matplotlib
```

```
Requirement already satisfied: matplotlib in C:\Users\Santiago\Desktop\Escuela\5to S
emestre ISSC\BigData\U1_1\.venv\Lib\site-packages (3.11.1)
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\Santiago\Desktop\Escuela
\5to Semestre ISSC\BigData\U1_1\.venv\Lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in C:\Users\Santiago\Desktop\Escuela\5to
Semestre ISSC\BigData\U1_1\.venv\Lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in C:\Users\Santiago\Desktop\Escuel
a\5to Semestre ISSC\BigData\U1_1\.venv\Lib\site-packages (from matplotlib) (4.64.0)
Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\Santiago\Desktop\Escuel
a\5to Semestre ISSC\BigData\U1_1\.venv\Lib\site-packages (from matplotlib) (1.5.1)
Requirement already satisfied: numpy>=1.25 in C:\Users\Santiago\Desktop\Escuela\5to
Semestre ISSC\BigData\U1_1\.venv\Lib\site-packages (from matplotlib) (2.5.2)
Requirement already satisfied: packaging>=20.0 in C:\Users\Santiago\Desktop\Escuela
\5to Semestre ISSC\BigData\U1_1\.venv\Lib\site-packages (from matplotlib) (26.3)
Requirement already satisfied: pillow>=9 in C:\Users\Santiago\Desktop\Escuela\5to Se
mestre ISSC\BigData\U1_1\.venv\Lib\site-packages (from matplotlib) (12.3.0)
Requirement already satisfied: pyparsing>=3 in C:\Users\Santiago\Desktop\Escuela\5to
Semestre ISSC\BigData\U1_1\.venv\Lib\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\Santiago\Desktop\Esc
uela\5to Semestre ISSC\BigData\U1_1\.venv\Lib\site-packages (from matplotlib) (2.9.
0.post0)
Requirement already satisfied: six>=1.5 in C:\Users\Santiago\Desktop\Escuela\5to Sem
estre ISSC\BigData\U1_1\.venv\Lib\site-packages (from python-dateutil>=2.7->matplotl
ib) (1.17.0)
```

In [35]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import requests

# Fetch the data.
df = pd.read_csv("https://ourworldindata.org/grapher/children-born-per-woman.csv?v=

# Fetch the metadata
metadata = requests.get("https://ourworldindata.org/grapher/children-born-per-woman
```

In [36]:
```
df.head()
```

Out[36]:

	entity	code	year	fertility_rate_hist
0	Afghanistan	AFG	1950	7.248
1	Afghanistan	AFG	1951	7.260
2	Afghanistan	AFG	1952	7.260
3	Afghanistan	AFG	1953	7.266
4	Afghanistan	AFG	1954	7.254

In [37]:
```
df.shape
```

Out[37]:
```
(19402, 4)
```

In [38]:
```
df.columns.tolist()
```

Out[38]:
```
['entity', 'code', 'year', 'fertility_rate_hist']
```

In [39]:
```
df.dtypes
```

Out[39]:
```
entity                   str
code                     str
year                   int64
fertility_rate_hist    float64
dtype: object
```

In [40]:
```
mexico = df[df['entity'] == 'Mexico']
japan = df[df['entity'] == 'Japan']
```

In [41]:
```
mexico.shape
```

Out[41]:
```
(74, 4)
```

In [42]:
```
japan.shape
```

Out[42]:
```
(77, 4)
```

In [43]:
```
year_mexico = mexico['year'].to_numpy()
rate_mexico = mexico['fertility_rate_hist'].to_numpy()

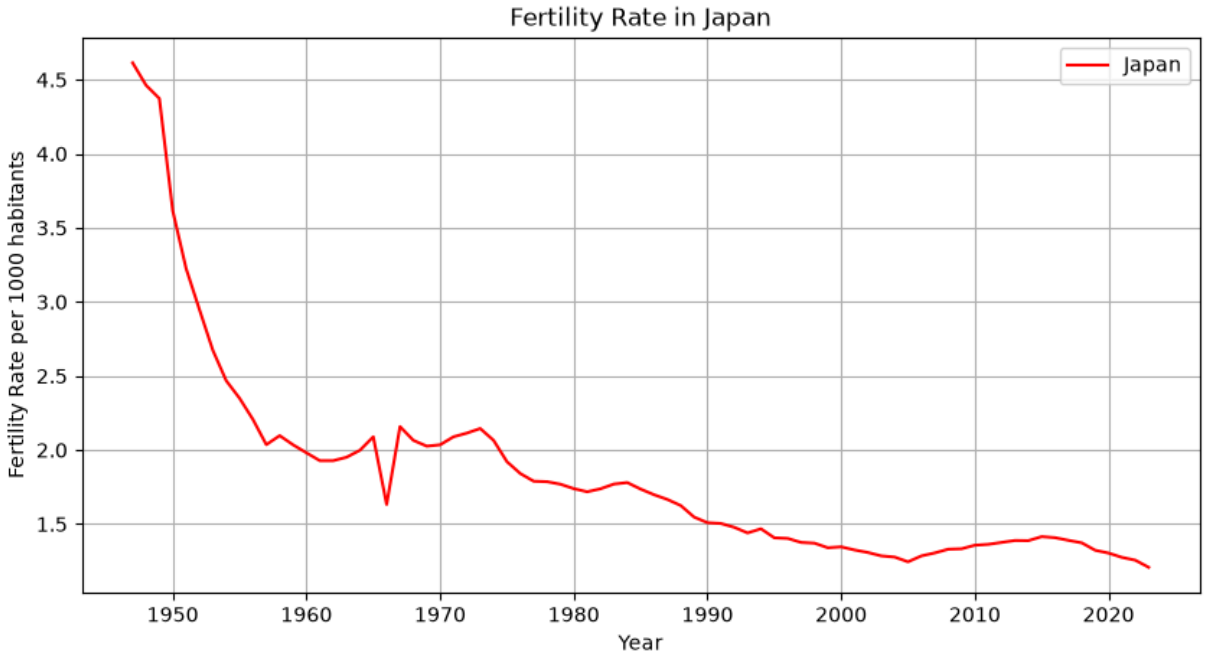
year_japan = japan['year'].to_numpy()
rate_japan = japan['fertility_rate_hist'].to_numpy()
```

In [44]:
```
promedio_mexico = np.mean(rate_mexico)
```

In [45]:
```
plt.figure(figsize=(10, 5))
plt.plot(
    year_japan, rate_japan, label='Japan', color='red'
)

plt.title('Fertility Rate in Japan')
plt.xlabel('Year')
plt.ylabel('Fertility Rate per 1000 habitants')
plt.grid()
plt.legend()
```

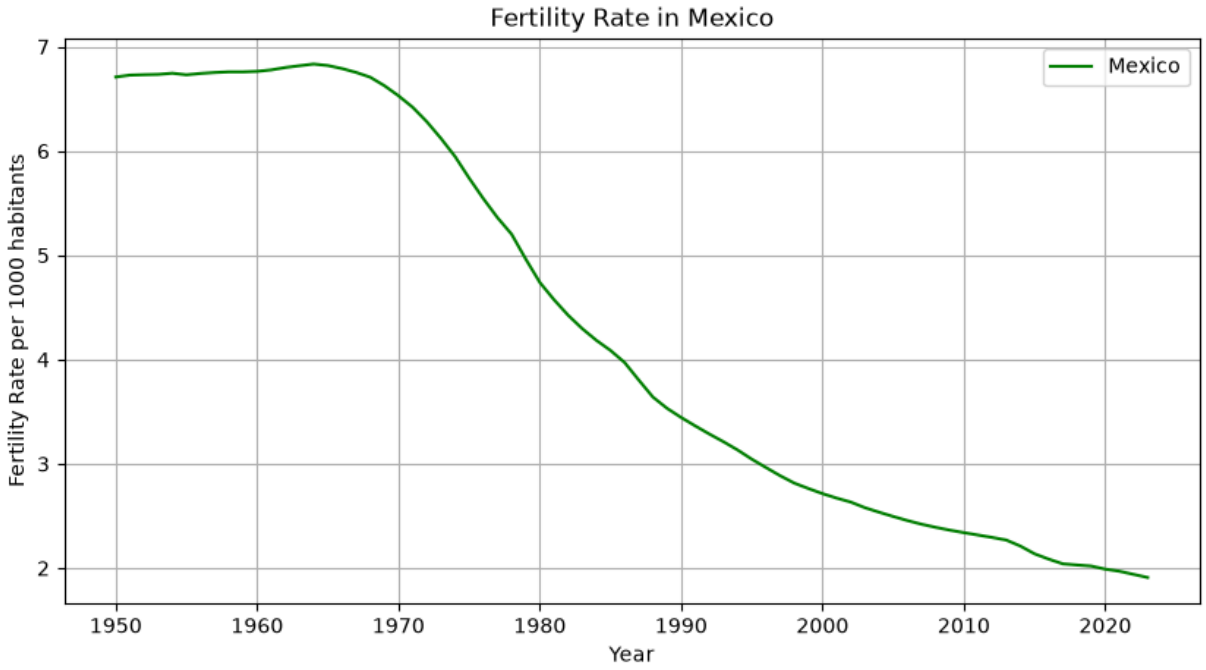
Out[45]:
```
<matplotlib.legend.Legend at 0x1c1fad95fd0>
```



In [46]:
```
plt.figure(figsize=(10, 5))
plt.plot(
    year_mexico, rate_mexico, label='Mexico', color='green'
)

plt.title('Fertility Rate in Mexico')
plt.xlabel('Year')
plt.ylabel('Fertility Rate per 1000 habitants')
plt.grid()
plt.legend()
```

Out[46]: <matplotlib.legend.Legend at 0x1c1fadff770>

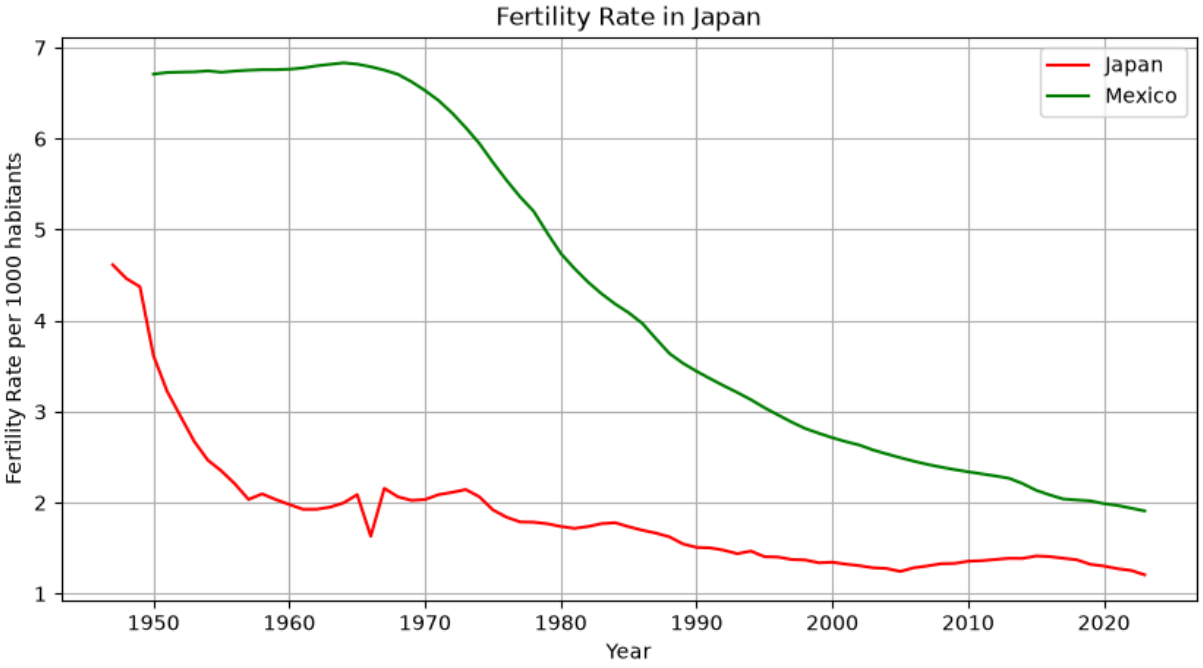


In [47]:
```
plt.figure(figsize=(10, 5))
plt.plot(
    year_japan, rate_japan, label='Japan', color='red'
)

plt.plot(
    year_mexico, rate_mexico, label='Mexico', color='green'
)

plt.title('Fertility Rate in Japan')
plt.xlabel('Year')
plt.ylabel('Fertility Rate per 1000 habitants')
plt.grid()
plt.legend()
```

Out[47]: <matplotlib.legend.Legend at 0x1c1fb240c20>



In [48]:
```
mexico.head(5)
```

Out[48]:

	entity	code	year	fertility_rate_hist
11125	Mexico	MEX	1950	6.709
11126	Mexico	MEX	1951	6.727
11127	Mexico	MEX	1952	6.731
11128	Mexico	MEX	1953	6.734
11129	Mexico	MEX	1954	6.745

In [49]:
```
japan.head(5)
```

Out[49]:

	entity	code	year	fertility_rate_hist
8384	Japan	JPN	1947	4.614
8385	Japan	JPN	1948	4.463
8386	Japan	JPN	1949	4.373
8387	Japan	JPN	1950	3.615
8388	Japan	JPN	1951	3.225

In [51]:
```
year_mexico.shape
```

Out[51]:
```
(74,)
```

In [52]:
```
year_japan.shape
```

Out[52]:
```
(77,)
```

In [53]:
```
mascara_coincidencia_year = year_japan >= year_mexico.min()
mascara_coincidencia_year
```

Out[53]:
```
array([False, False, False,  True,  True,  True,  True,  True,  True,
        True,  True,  True,  True,  True,  True,  True,  True,  True,
        True,  True,  True,  True,  True,  True,  True,  True,  True,
        True,  True,  True,  True,  True,  True,  True,  True,  True,
        True,  True,  True,  True,  True,  True,  True,  True,  True,
        True,  True,  True,  True,  True,  True,  True,  True,  True,
        True,  True,  True,  True,  True,  True,  True,  True,  True,
        True,  True,  True,  True,  True])
```

In [54]:
```
year_japan2 = rate_japan[mascara_coincidencia_year]
```

In [55]:
```
rate_japan2 = rate_japan[mascara_coincidencia_year]
rate_japan2.shape
```

Out[55]:
```
(74,)
```

In [57]:
```
len(rate_japan2)
```

Out[57]:
```
74
```

In [58]:
```
diferencia_jp_mx = rate_japan2 - rate_mexico
```

In [60]:
```
plt.figure(figsize=(10, 5))

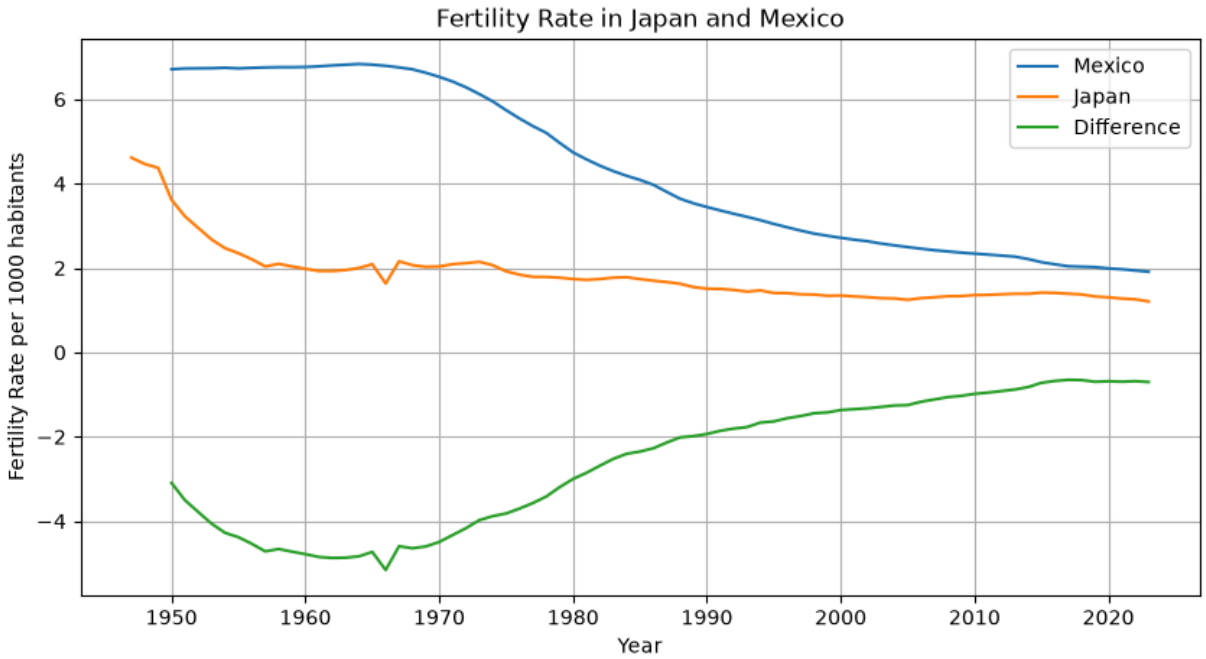
plt.plot(
    year_mexico,
    rate_mexico,
    label='Mexico',
)
```

```
plt.plot(
    year_japan,
    rate_japan,
    label='Japan',
)

plt.plot(
    year_mexico,
    diferencia_jp_mx,
    label='Difference',
)

plt.title('Fertility Rate in Japan and Mexico')
plt.xlabel('Year')
plt.ylabel('Fertility Rate per 1000 habitants')
plt.grid()
plt.legend()
```

Out[60]: <matplotlib.legend.Legend at 0x1c1fb510440>



In [61]:
```
common_years = np.intersect1d(year_mexico, year_japan)
```

In [62]:
```
len(common_years)
```

Out[62]: 74

In [63]:
```
type(common_years)
```

Out[63]: numpy.ndarray

In [64]:
```
mexico_common_years = mexico[mexico['year'].isin(common_years)]
japan_common_years = japan[japan['year'].isin(common_years)]
```

In [65]:
```
mexico_rate_common_years = mexico_common_years['fertility_rate_hist'].isin(common_y
japan_rate_common_years = japan_common_years['fertility_rate_hist'].isin(common_yea
```

In [66]:
```
mex_common_years = mexico_common_years['year'].to_numpy()
jap_common_years = japan_common_years['year'].to_numpy()

mex_rate_common_years = mexico_common_years['fertility_rate_hist'].to_numpy()
jap_rate_common_years = japan_common_years['fertility_rate_hist'].to_numpy()
```

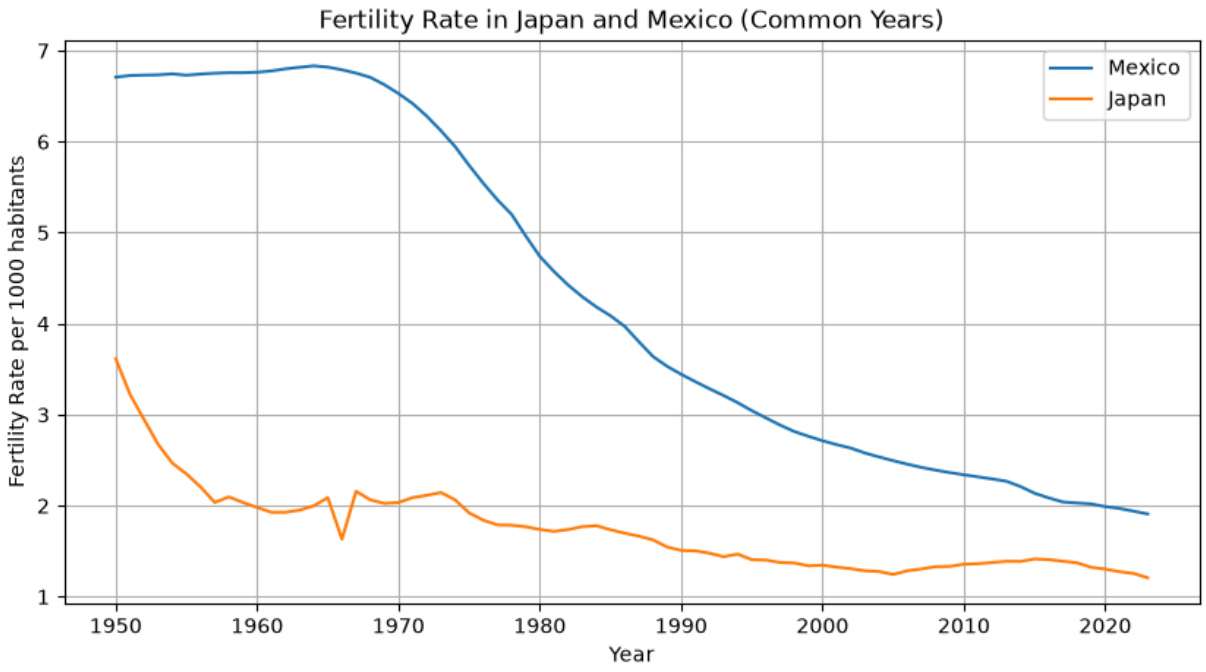
In [67]:
```
print(year_japan[30])
```

```
1977
```

In [68]:
```
plt.figure(figsize=(10, 5))

plt.plot(
    mex_common_years,
    mex_rate_common_years,
    label='Mexico',
)
plt.plot(
    jap_common_years,
    jap_rate_common_years,
    label='Japan',
)
plt.title('Fertility Rate in Japan and Mexico (Common Years)')
plt.xlabel('Year')
plt.ylabel('Fertility Rate per 1000 habitants')
plt.grid()
plt.legend()
```

Out[68]:
```
<matplotlib.legend.Legend at 0x1c1fb55dfd0>
```



In [69]:
```
comparison_mex_jap = pd.merge(
    mexico[["year", "fertility_rate_hist"]],
    japan[["year", "fertility_rate_hist"]],
    on="year",
    suffixes=("_mex", "_jap")
    )
```

In [70]:
```
comparison_mex_jap.head(3)
```

Out[70]:

	year	fertility_rate_hist_mex	fertility_rate_hist_jap
0	1950	6.709	3.615
1	1951	6.727	3.225
2	1952	6.731	2.947

In [71]:
```
comparison_mex_jap["diferencia"] = (
    comparison_mex_jap["fertility_rate_hist_jap"] - comparison_mex_jap["fertility_r
)
```

In [72]:
```
comparison_mex_jap.head(3)
```

Out[72]:

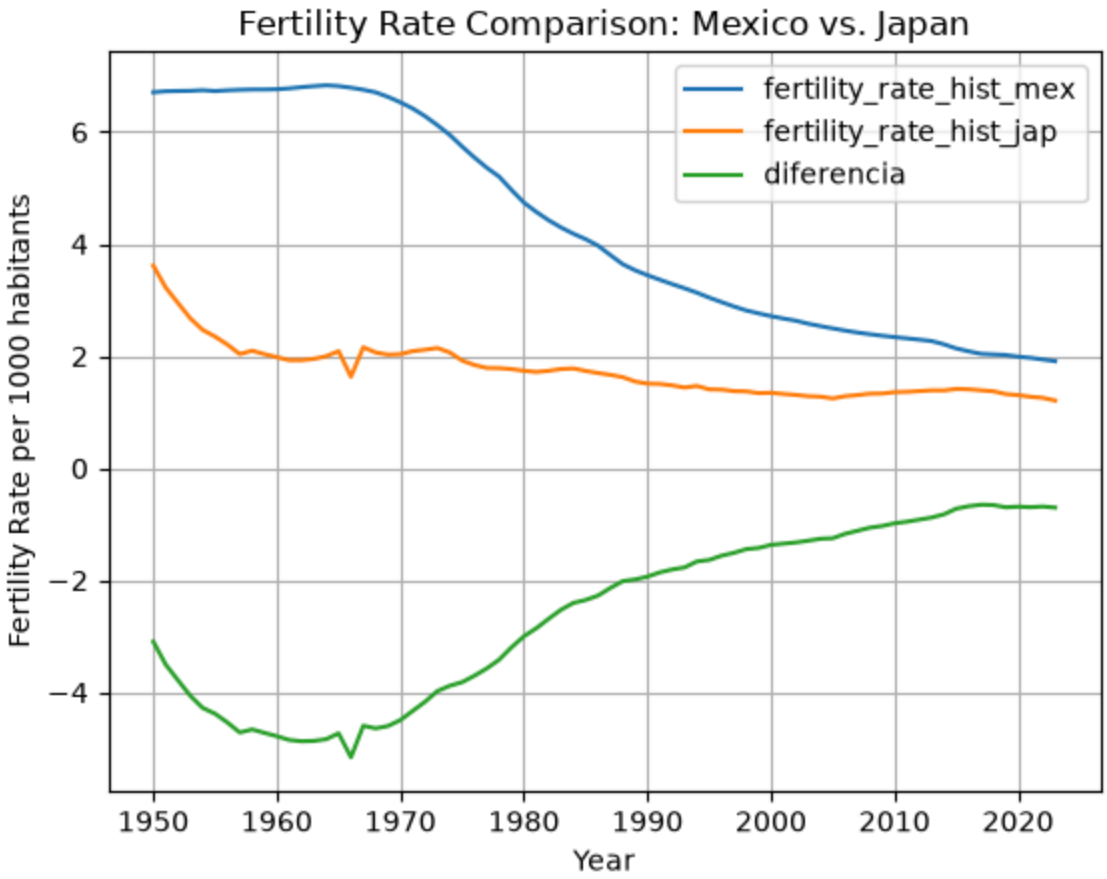
	year	fertility_rate_hist_mex	fertility_rate_hist_jap	diferencia
0	1950	6.709	3.615	-3.094
1	1951	6.727	3.225	-3.502
2	1952	6.731	2.947	-3.784

In [73]:
```
plt.figure(figsize=(10, 5))

comparison_mex_jap.plot(
    x = "year",
    y = ["fertility_rate_hist_mex", "fertility_rate_hist_jap", "diferencia"],
    kind = "line"
)

plt.title("Fertility Rate Comparison: Mexico vs. Japan")
plt.xlabel("Year")
plt.ylabel("Fertility Rate per 1000 habitants")
plt.grid()
plt.legend()
```

Out[73]:
```
<matplotlib.legend.Legend at 0x1c1fb976ba0>
<Figure size 1000x500 with 0 Axes>
```


Fertility Rate Comparison: Mexico vs. Japan
fertility_rate_hist_mex
fertility_rate_hist_jap
diferencia
Fertility Rate per 1000 habitants
6
4
2
0
−2
−4
1950
1960
1970
1980
1990
2000
2010
2020
Year